

Simple DC TO DC CONVERTER For MICROCONTROLLER KIT

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This article describes a simple, low-cost DC to DC converter for a microcontroller (MCU) kit. This converter is built around popular NE555 timer. It can provide positive and negative output voltages simultaneously. Output currents are limited only by the output capabilities of NE555.

Most MCU kits on the market are available with a single power supply, mostly +5V DC. Also, the transformer with or without a rectifier is usually external, for example, in the form of a wall adaptor. But for many experiments and real-world projects, we need power supplies that provide a bit higher voltages than +5V and a bit lower than the ground such as +7.5V and -2V, respectively.

You may use an external power supply for providing these voltages. But since the required currents are usually a few milliamperes, you can use a low-power DC to DC converter to produce the needed voltages. Power supply for the DC to DC converter can come before or after the regulator for +5V. But the produced output voltages (+V_{OUT} and -V_{OUT}) will depend on input voltage.

Circuit and working

Fig. 1 shows the circuit diagram of the DC to DC converter (MCU kit companion board) built around NE555 (IC1). The board is also called DC-DC companion board because it is complementary to the MCU kit for many applications. NE555 works as

a square-wave oscillator.

Frequencies depend mainly on resistor R1 and capacitor C1, and can be selected up to around 200kHz. With R1 as 10-kilo-ohm and C1 as 10nF (0.01µF), 5.1nF, 3.3nF and 330pF, we have frequencies around 7kHz, 14kHz, 20kHz and 150kHz.

Capacitors C4, C5 and C6, and diodes D1 and D2 produce positive output voltage (+V_{OUT}). With a 9V input power supply (+V_{EE} = +9V) the circuit produces +V_{OUT} around +16V without any load, +14.5V with 15mA load and +13V with load of 20mA. This is enough for giving power to several operational amplifiers and comparators that

have low power consumption.

Diodes D3 and D4, and capacitors C7, C8 and C10 produce the negative output voltage (-V_{OUT}). With +V_{EE} = 9V power supply, the circuit produces -V_{OUT} around -7V without any load, -5.3V with load of 16mA and -3.7V with load of 35mA. This is good enough for many analogue and interface applications.

Resistor R3 and connector CON2

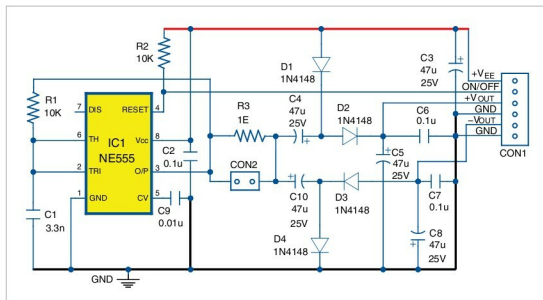


Fig. 1: Circuit diagram of the DC to DC converter for the MCU kit

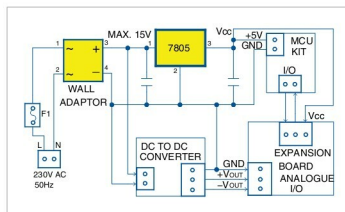


Fig. 2: Connection of the DC to DC converter to a wall adaptor